Information-Theoretic SCX
——rate-distortion

Information-Theoretic SCX: Rate-Distortion Theory of Multi-Expert Information Fusion

SCX

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Abstract

SCXState-Conditioned eXpertise$M$ howeverSCXtheoremTheorem 1–4Core communicationupper boundencodingcompression-channelencoding theorem——rate-distortionSlepian-WolfencodingShannontheorem—— SCX

rate-distortionSCXrate-distortionVersion $Q(\mathbf{V})\mathbf{V} \in \{0,1\}^M$communicationStatus$R$ communicationSCXlower bound $P_e \geq (H(Y) - R - \log 2)/\log|\mathcal{Y}|$ **-communication-** Bottleneck-compression

Slepian-Wolfencoding$M$source ProofSlepian-WolftheoremSCXVersion$M$compression $\sum_{m\in\mathcal{T}} R_m \geq H(\mathbf{V}_{\mathcal{T}} \mid \mathbf{V}_{\mathcal{T}^c})$ all$\mathcal{T}$$H(\mathbf{V})$$M \cdot H_2(p_{\text{clean}})$ encoding—— SCX[**Honest Strike**]

ShannontheoremSCXProofSCXthere exists iencodingcompressionScoreii channelencodingif and only ifsatisfying otherwise-channelencoding

Theorem 2Theorem 2Fanoawaiting rate-distortionBottleneck $\delta(R) = \inf_{P_{U|X}:I(X;U)\leq R} I(Y;U)$Proof$R$ awaitingTheorem 2upper bound $F_{1,\text{SCX}} \leq F_{1,\text{base}} + C_F \cdot r(\delta(0))$ where$r(\cdot)$$XY$rate-distortion

Honest Strike Rigorous Prooftheoremassumption SCX

- rate-distortionCompactnessi.i.d.

Assumption 0.1. *——Statusthere exists*

Slepian-Wolfencodingfeasibilityencodingthere exists common randomness——SCX

theoremSCX$M \to \infty$, $n \to \infty$

Theorem 2although$\delta(R)$ Theorem 2——there exists-

KeywordsSCXrate-distortionSlepian-WolfencodingShannontheoremBottleneck Mutual InformationFanoawaiting

1 SCX

1.1 SCX

SCXMNStatus Coretheorem[11]

- **Theorem 1**Consistency$F_1 \geq 1 - \frac{1}{\eta}\sum_s \rho_s \exp(-2M\Delta_s^2)$
- **Theorem 2**$F_{1,\text{SCX}} \leq F_{1,\text{base}} + C_F\sqrt{\delta/2}$
- **Theorem 3**-there existsawaiting
- **Theorem 4**MinimaxChernoff-Steinlower bound

howevertheoremthere exists Theorem 1Chernoff-HoeffdingawaitingTheorem 2Fanoawaiting Theorem 4Chernoff-Steinlemma—— Core——rate-distortionsourceencodingtheoremBottleneck——SCX

CoreSCX M**sourcedestination** communicationencoding **channel**Perspectivetheoremassumption

1.2 SCX

Definition 1.1 (SCX)**.** *SCXdefine*

$$\boxed{\mathcal{T}_{SCX} = (\mathcal{S}, \mathcal{E}, \mathcal{C}, \mathcal{A}, \mathcal{D})} \tag{1}$$

where

- $\mathcal{S}$*StatusN*$Y_s \in \mathcal{Y}$
- $\mathcal{E} = \{E_1, \ldots, E_M\}$*each*$E_m$*Statuss* $V_m(s) = \mathbf{1}[E_m(\phi(s)) \neq y_s] \in \{0, 1\}$
- $\mathcal{C}$*communicationchannel*$\mathbf{V}(s) = (V_1(s), \ldots, V_M(s))$ *compression*$Q(\mathbf{V}(s)) \in \{1, \ldots, 2^R\}$ *satisfying*$R = \frac{1}{N}\sum_s \log |Q(\mathbf{V}(s))|$
- $\mathcal{A}$*compression* $\hat{Y}_s = \mathcal{A}(Q(\mathbf{V}(s)))$
- $\mathcal{D}d(\hat{Y}_s, Y_s) = \mathbf{1}[\hat{Y}_s \neq Y_s]$

SCX**-channelencoding** Msource$\to$ encodingcompression/$\to$ channelcommunication$\to$ ——rate-distortionSlepian-Wolftheorem——SCX

Definition 1.2. *Statussy*$_s M\mathbf{V} \in \{0, 1\}^M$

$$P(\mathbf{V} = \mathbf{v} \mid Y = y) = \prod_{m=1}^{M} p_y^{v_m}(1 - p_y)^{1-v_m} \quad (assumption) \tag{2}$$

where$p_{clean,s} = P(V_m = 1 \mid Y = clean)$ $p_{noisy,s} = P(V_m = 1 \mid Y = noisy)$
assumptiondefine1.1SCX assumption§3§5

[Honest Strike] Honest Strike SCXCoreiassumptionStatussatisfying asymptotic—— iiassumptioncommunication——Bottleneck asymptotic

2 rate-distortioncommunicationSCXlower bound

2.1 rate-distortionSCX

rate-distortionRate-Distortion Theorysourcecompression Core[4]source$X \sim P_X d(x, \hat{x})$ rate-distortiondefine

$$R(D) = \min_{P_{\hat{X}|X}:\mathbb{E}[d(X,\hat{X})] \leq D} I(X; \hat{X}) \tag{3}$$

SCX"source"$M\mathbf{V} \in \{0, 1\}^M$ "" $P_e = \mathbb{E}[d(\hat{Y}, Y)] = P(\hat{Y} \neq Y)$ ""Statuscommunication$R$
Core**communication**R**SCXlower bound**

2.2 theoremrate-distortionlower bound

Theorem 2.1 (SCXrate-distortionlower bound——)**.** M*Statuss*$\mathbf{V} \in \{0, 1\}^M Y$ $\mathbf{V}$ *Version*$Q(\mathbf{V})$*satisfying* $I(\mathbf{V}; Q(\mathbf{V})) \leq R$*Statussatisfying*

$$\boxed{P_e \geq \frac{H(Y) - R - \log 2}{\log |\mathcal{Y}|}} \tag{4}$$

awaitingε*communication*

$$\boxed{R_{\min}(\varepsilon) \geq H(Y) - \log 2 - \varepsilon \log |\mathcal{Y}|} \tag{5}$$

Proof. **1FanoawaitingFanoawaiting**[4]

$$H(Y \mid Q(\mathbf{V})) \leq H_2(P_e) + P_e \cdot \log(|\mathcal{Y}| - 1) \leq \log 2 + P_e \log |\mathcal{Y}| \tag{6}$$

where$H_2(P_e) \leq \log 2$Entropyupper bound

2Entropyawaiting

$$\begin{aligned} H(Y \mid Q(\mathbf{V})) &= H(Y) - I(Y; Q(\mathbf{V})) && (7)\\ &\geq H(Y) - I(\mathbf{V}; Q(\mathbf{V})) \quad \text{DPI: } Y \to \mathbf{V} \to Q(\mathbf{V}) && (8)\\ &\geq H(Y) - R && (9) \end{aligned}$$

3lower bound

$$\begin{aligned} H(Y) - R \leq H(Y \mid Q(\mathbf{V})) &\leq \log 2 + P_e \log |\mathcal{Y}| && (10)\\ \implies P_e &\geq \frac{H(Y) - R - \log 2}{\log |\mathcal{Y}|} && (11) \end{aligned}$$

4Compactnesslower boundasymptotic i$Q(\mathbf{V})Y$sufficient statistici.e.$Y \perp\!\!\!\perp \mathbf{V} \mid Q(\mathbf{V})$ iiFanoawaitingawaitingall iiiawaitingi.e.$I(\mathbf{V}; Q(\mathbf{V})) = R$ $M \to \infty$awaiting □ □

[Rigorous Proof] ProofStatus:1–3Fanoawaiting+DPI+Entropy 4Compactnessasymptotic——

Remark 2.2 (Theorem 1——Chernoff vs. rate-distortion). Theorem 1F_1lower boundChernoff-Hoeffdingtheorem2.1 $P_e$lower boundrate-distortion

- Theorem 1M——$M\Delta_s > 0$ $F_1$1“”
- theorem2.1**communication**——M communication$R$lower bound“”

Theorem 1assumptioncommunicationall theorem2.1assumptionchannel rate-distortion

2.3 rate-distortion

Corollary 2.3 (-communication-)**.** *SCXthere exists*

$$\boxed{\underbrace{R}_{\textit{communication}} \cdot \underbrace{N}_{\textit{Status}} \geq H(\mathbf{V}_1, \ldots, \mathbf{V}_M) - \underbrace{N \cdot H_2(P_e)}} \tag{12}$$

$MH(\mathbf{V})$encodingsimultaneouslyΔ_sGrowth $P_eR$there existsM^*

Proof. rate-distortiondefinetheorem2.1$H(\mathbf{V})M$Growth Growth$MP_eM\exp(-2M\Delta_s^2)$Theorem 1 M^* □ □

[Heuristic] ProofStatus:Heuristic$M^*H(\mathbf{V})P_e(M)$ $MH(\mathbf{V})$ ——SCX

2.4 BottleneckSCXcompression

rate-distortionlower boundcompressionBottleneck Information Bottleneck, IB[5]

Definition 2.4 (SCXBottleneck)**.** *SCXBottleneck*

$$\boxed{\min_{P_{Q|\mathbf{V}}} \; I(\mathbf{V}; Q) - \beta \cdot I(Y; Q)} \tag{13}$$

where$Q$compression$\beta > 0$compression $\beta \to 0$compression$Q\mathbf{V}$ $\beta \to \infty Q$all$Y$

Proposition 2.5 (SCX-IBcompression)**.** *source*$\{0,1\}^M$*SCX-IBsatisfying*

$$P_{Q|V}(q \mid \mathbf{v}) = \frac{P_Q(q)}{Z(\mathbf{v},\beta)} \exp\big(-\beta \cdot \mathrm{D}_{\mathrm{KL}}(P_{Y|\mathbf{V}=\mathbf{v}} \| P_{Y|Q=q})\big) \tag{14}$$

where$Z(\mathbf{v},\beta)$ *i*$P_{Q|V}P_{Y|Q}$ *ii*$P_{Y|Q}P_{Q|V}$

Proof. Tishbyawaiting[5]CoreSCXsource source$X\mathbf{V}Y$IB convexity$P_{Q|V}P_{Y|Q}$ ☐ ☐

[Rigorous Proof] ProofStatus:IBconvergence SCX$P_{Y|\mathbf{V}}$ IBcompression

[Honest Strike] Honest StrikeIBfeasibility althoughIBSCX2^M—— $M = 10$i.e.1024Status$M = 20$i.e. forawaiting IBIB[6]

2.5 Masymptoticrate-distortion

Theorem 2.6 (asymptotic——Strassen)**.** *for*MN*Statuscommunication*NRD *there existsif and only ifthere exists*$P_{Q|\mathbf{V}}$*such that*

$$\boxed{\frac{1}{N}\sum_{s=1}^{N} I(\mathbf{V}_s;Q_s) \le R \qquad \frac{1}{N}\sum_{s=1}^{N} \mathbb{E}[d(\hat{Y}_s,Y_s)] \le D} \tag{15}$$

N*there exists**channel**channel dispersion* $O(1/\sqrt{N})$*such thatasymptotic* $\sqrt{V(D)/N}\cdot Q^{-1}(\epsilon)$*where*$V(D)$*rate-distortion* Q^{-1}

Proof. Kostina & Verdú[7]asymptoticrate-distortionSCX asymptoticBerry-Esseen ☐ ☐

[Rigorous Proof] ProofStatus:channelencodingStrassen 1962Hayashi2006$V(D)$source——SCXall$p_{\mathrm{clean},s}p_{\mathrm{noisy},s}$

forSCX$N \approx 10^3$–10^6Status asymptotic10%–30% communication

3 Slepian-Wolfencoding$M$compression

3.1 encodingsourceencoding

SCXcommunication$M$—— eachencodingencodingcommunication sourceencodingDistributed Source Coding

Definition 3.1 (SCXencoding)**.** M*encoding*$V_m \in \{0,1\}$ *encoding*$m$$C_m(V_m) \in \{0,1\}^{nR_m}$ *all*$(C_1,\ldots,C_M)$ $\hat{\mathbf{V}} = (\hat{V}_1,\ldots,\hat{V}_M)$ $\lim_{n\to\infty} P(\hat{\mathbf{V}} \neq \mathbf{V}) = 0$ n*Status0*

3.2 Slepian-WolftheoremSCXVersion

Theorem 3.2 (SCX Slepian-Wolfencodingtheorem——compression)**.** $M\mathbf{V} = (V_1,\ldots,V_M)P_{\mathbf{V}}$ *there existsencodingany*$\mathbf{V}$*if and only if*$(R_1,\ldots,R_M)$*satisfying*

$$\boxed{\sum_{m\in\mathcal{T}} R_m \ge H(\mathbf{V}_{\mathcal{T}} \mid \mathbf{V}_{\mathcal{T}^c}), \quad \forall \mathcal{T} \subseteq \{1,\ldots,M\}} \tag{16}$$

where$\mathbf{V}_{\mathcal{T}} = \{V_m : m \in \mathcal{T}\}$ $\mathbf{V}_{\mathcal{T}^c} = \{V_m : m \notin \mathcal{T}\}$ ***in particular****satisfying*

$$\boxed{\sum_{m=1}^{M} R_m \ge H(\mathbf{V})} \tag{17}$$

$H(\mathbf{V}) = \sum_m H(V_m) = M \cdot H_2(p)$ *Slepian-Wolfencoding* $\mathrm{Cov}(V_i,V_j) \to \mathrm{Var}(V_i)$ $H(\mathbf{V}) \ll \sum_m H(V_m)$*——encoding*

Proof. **1**random binning eachencoding$m\mathbf{v}_m^n 2^{nR_m}$bin all lemma1

2theorem$\mathcal{T}$awaitingEntropy compression$\geq 1/2$——Fanoawaitinglower bound

3V_m $H(\mathbf{V}) = \sum_m H(V_m) = \sum_m H_2(p_{\text{clean},s})$ allSlepian-Wolf$R_m \geq H(V_m)$——encoding □

□

[Rigorous Proof] ProofStatus:theorem3.2 Slepian-Wolftheorem[2]SCXsource SCXSlepian-Wolf——becausesource $\{0,1\}^M 2^M$

3.3 ——

Theorem 3.3 (——Slepian-WolfSCX). *$\rho_{ij} = (V_i, V_j)ij$ Slepian-Wolfcompressionforencoding*

$$\boxed{G_{SW}(\boldsymbol{\rho}) = \frac{\sum_{m=1}^{M} H(V_m)}{H(\mathbf{V})} - 1 \geq 0} \tag{18}$$

*$G_{SW}\rho_{ij} \to 1 H(\mathbf{V}) \to H(V_1)$ 1encodingallM Theorem 1Δ_s M_{eff}0——**compressionSCX***

Proof. **1Entropy**Entropy $H(\mathbf{V}) = \sum_{m=1}^{M} H(V_m \mid V_1, \ldots, V_{m-1}) \leq \sum_{m=1}^{M} H(V_m)$ →Entropy$H(V_m \mid V_{<m})$→$H(\mathbf{V})$→G_{SW}

2theorem2.5.1M_{eff} $M_{\text{eff}} = M/(1 + (M-1)\bar{\rho})$ $\bar{\rho} \to 1 M_{\text{eff}} \to 1$——$\Delta_s M$Decays

3$G_{\text{SW}} M_{\text{eff}} \bar{\rho}$ $G_{\text{SW}} \uparrow \iff \bar{\rho} \uparrow \iff M_{\text{eff}} \downarrow$ **“”communicationreliability** □ □

[Honest Strike] Honest StrikeSlepian-WolfSCX Slepian-Wolfencoding——

(1) encoding$P(V_1, \ldots, V_M)$—— SCX$p_{\text{clean},s} p_{\text{noisy},s}$ encoding→Fails

(2) allencoding—— communication

(3) Slepian-Wolfasymptoticoptimality$n \to \infty$“encoding”—— SCX$n = 1$awaiting for$n = 1$Slepian-Wolfencoding

Corollary 3.4 (encodingSlepian-Wolf). *$n = 1$Slepian-Wolfencoding**encoding***

$$\boxed{R_m \geq H(V_m) = H_2(p_{clean,s}) \;\; H_2(p_{noisy,s})} \tag{19}$$

encoding$n \gg 1$batch auditing

[Open Problem] Open Problem1Slepian-Wolf there existsSCXzero-delay, $n = 1$encoding i.e. ——Slepian-Wolftheorem—— SCX$[0,1]$

3.4 assumptionSlepian-Wolf

satisfyingSCXSlepian-Wolf $R_m \geq H_2(p_{\text{clean},s})$ there exists because$H(V_i \mid V_j) \leq H(V_i)$i.e.

SCX“”communication →Slepian-Wolf→$H(V_m)$→communication$M \cdot H_2(p)$ SCXConclusion——reliabilitycommunication

4 ShannontheoremSCXencodingchannelencoding

4.1 theoremSCX

Shannon**theorem**Separation Theorem[1] communicationsourceencodingcompressionChannel Capacitychannelencoding channeloptimality theoremsuch thatcommunication——compression

SCXthere exists

- **SCX“encoding”**Mcompression **Score**$Q(s)$—— “compression”2^MStatuscompression$Q$
- **SCX“channelencoding”**/Verdict —— “”

SCXencodingchannelencoding

4.2 SCXtheorem

Theorem 4.1 (SCXtheorem——)**.** *$M$$\mathbf{V}$$Y$ SCXthere exists*

(i) ***encoding**$\mathbf{V}$ compressionsufficient statistic $T(\mathbf{V}) = \sum_{m=1}^{M} w_m V_m$ where$w_m^* = \log \frac{p_{noisy,m}(1-p_{clean,m})}{p_{clean,m}(1-p_{noisy,m})}$*

(ii) ***channelencoding**$T(\mathbf{V})\tau$ Verdict$Y$*

F_1Score

Proof. **1sufficient statistic**assumption

$$\Lambda(\mathbf{V}) = \sum_{m=1}^{M} \log \frac{P(V_m \mid Y = \text{noisy})}{P(V_m \mid Y = \text{clean})} = \sum_{m=1}^{M} V_m \cdot w_m^* + \text{const} \tag{20}$$

Fisher-Neymantheorem$T(\mathbf{V}) = \sum_m w_m^* V_m Y$**sufficient statistic**

2optimalitysufficient statistic$Y \perp\!\!\!\perp \mathbf{V} \mid T(\mathbf{V})$ therefore$\mathbf{V}T(\mathbf{V})$ sourceencoding$\mathbf{V} \to T(\mathbf{V})Y$ channelencoding$T(\mathbf{V})$

3assumption$T(\mathbf{V}) = \sum_m w_m V_m$ sufficient statistic——Y -channelencoding □ □

[Rigorous Proof] ProofStatus:1–2sufficient statistic 3awaitingConclusion if and only ifMarkov$Y \to T(\mathbf{V}) \to \mathbf{V}T(\mathbf{V})$

Theorem 4.2 (theorem——encoding)**.** *there exists*$\mathrm{Cov}(V_i, V_j) \neq 0$ *-channelencoding*

$$\boxed{F_1^{joint} \geq F_1^{separate} + \underbrace{\frac{I(Y;\mathbf{V}) - I(Y;T(\mathbf{V}))}{\log|\mathcal{Y}|}} - O\left(\frac{1}{\sqrt{M}}\right)} \tag{21}$$

Y

Proof. awaitingFanoawaiting $T(\mathbf{V})$sufficient statistic $\mathbf{V}$ Mutual Information$I(Y;\mathbf{V}) - I(Y;T(\mathbf{V}))$lower bound Cover & Thomas[4]7FanoDPI □ □

[Rigorous Proof] ProofStatus:$I(Y;\mathbf{V}) - I(Y;T(\mathbf{V}))$ $\mathbf{V}$——

4.3 theorem vs.

Corollary 4.3 (SCX)**.** *(1) → Verdict*

(2) ***there exists***

(3) lower bound —— Overfitting

[Honest Strike] Honest Striketheoremasymptotic Shannontheorem**asymptotic**$n \to \infty$ SCX$n = 1n$due to isufficient statistic$T(\mathbf{V})$sufficiencyasymptotic—— $MT(\mathbf{V})Y$ iiassumptionMp-hacking

asymptotictheorem[8] SCXchannelencoding

5 Theorem 2Fanorate-distortionBottleneck

5.1 Theorem 2

Theorem 2[11]

$$F_{1,\text{SCX}} \leq F_{1,\text{base}} + C_F \cdot \sqrt{\frac{\delta}{2}} \tag{22}$$

where$\delta = I(Y; X_{\text{ideal}}) - I(Y; X)$ Mutual InformationC_F Theorem 2′define3.1ε_{PE} encoding$\varepsilon_{\text{PE}} = I(Y; P \mid X) - I(Y; \text{PE}(P) \mid X)$ upper bound

$$F_{1,\text{SCX}}^{\text{Situs}} \leq F_{1,\text{base}} + C_F \cdot \sqrt{\frac{\delta + 2\varepsilon_{\text{PE}}/C_F^2}{2}} \tag{23}$$

howeverthere exists

(1) Fanoawaiting$P_e$lower bound $\eta < 0.5$——lower boundvacuous because$H_2(\eta) < \log 2(H_2(\eta) - \log 2) < 0$

(2) **rate-distortion**δ —— i.e.rate-distortion$R(D)$Bottleneck

(3) $\delta = I(Y; X_{\text{ideal}}) - I(Y; X)$ $P_{Y|X}$ δ

5.2 $\delta(R)$——rate-distortion

Definition 5.1 ($\delta(R)$)**.** *$X X_{ideal} Y$ define*

$$\boxed{\delta(R) = I(Y; X_{ideal}) - \max_{P_{U|X}: I(X;U) \leq R} I(Y; U)} \tag{24}$$

where$R \geq 0$encoding$X$ $\delta(0) = I(Y; X_{ideal}) - I(Y; X) = \delta_{original}$define $\delta(\infty) = 0$$RUX$ $\delta(R)$$R$convexity$P_{X,Y}$

Theorem 5.2 (Theorem 2rate-distortion——)**.** *SCXX compression$U$ satisfying$I(X;U) \leq R$ F_1Scoreupper bound*

$$\boxed{F_{1,SCX}(R) \leq F_{1,base} + C_F \cdot r(\delta(R))} \tag{25}$$

where$r(\delta) = \sqrt{\delta / (2 \log |\mathcal{Y}|)}$rate-distortion- in particular

(i) $R \to 0$$\delta(R) \to \delta(0) = \delta$upper boundTheorem 2

(ii) $R \to \infty$$\delta(R) \to 0$upper bound$F_{1,base}$

(iii) for$R$upper boundTheorem 2$R = 0$$R = \infty$

Proof. **1rate-distortionFano**compression$U$$\leq R$

$$P_e(R) \geq \frac{H(Y) - I(Y; U) - \log 2}{\log |\mathcal{Y}|} \tag{26}$$

2

$$\begin{aligned} I(Y; U) &= I(Y; X_{\text{ideal}}) - [I(Y; X_{\text{ideal}}) - I(Y; U)] &(27)\\ &\leq I(Y; X_{\text{ideal}}) - \delta(R) &(28)\\ &= H(Y) - H(Y \mid X_{\text{ideal}}) - \delta(R) &(29) \end{aligned}$$

3F_1 21$H(Y \mid X_{\text{ideal}}) = 0$$Y$define $P_e(R) \geq \delta(R)/\log |\mathcal{Y}| - O(1/\log |\mathcal{Y}|)$ $F_1$$P_e$Theorem 2$C_F$upper bound

4convexitymonotonicity$\delta(R)$Bottleneck $\max_{P_{U|X}: I(X;U) \leq R} I(Y; U)$$R$Tishbyawaiting[5] therefore$r(\delta(R))$$R$ bound$R = 0$$R = \infty$ □

[Rigorous Proof] ProofStatus:1–3awaitingawaiting4$\delta(R)$ $P_{X,Y}$**any**$P_{X,Y}$$\delta(R)$**IB forsourcelower boundfor/source**

5.3 BottleneckPerspective

Theorem 2BottleneckInformation Plane

Definition 5.3 (SCX). *SCX$I(X;U)$compression$I(Y;U)$define Bottleneck$I(Y;U)^*(R) = \max_{P_{U|X}:I(X;U)\leq R} I(Y$ compression$\delta(R)$ $I(Y;U) = H(Y)I(X;U)$any*

Proposition 5.4. *XforX_{ideal} IB*

$$\boxed{\Delta_{IB}(R) = I_{ideal}(R) - I_X(R)} \tag{30}$$

where$I_{ideal}(R)I_X(R)$Bottleneck $\Delta_{IB}(R) > 0$all$R$awaiting —— Theorem 2

[Heuristic] ProofStatus: Bottleneckfor $X \in \mathbb{R}^{768}$BERTIB Alemiawaiting[6]Consistency convergenceSCX

5.4 $\varepsilon_{\mathrm{PE}}$rate-distortion

$\varepsilon_{\mathrm{PE}}$encodingrate-distortion

Definition 5.5 (rate-distortionencoding).

$$\boxed{\varepsilon_{\mathrm{PE}}(R) = I(Y;P \mid X) - \max_{P_{U|\mathrm{PE}(P)}:I(\mathrm{PE}(P);U)\leq R} I(Y;U \mid X)} \tag{31}$$

$\varepsilon_{\mathrm{PE}}(0) = I(Y;P \mid X)$*encoding* $\varepsilon_{\mathrm{PE}}(\infty) = \varepsilon_{\mathrm{PE}}$*define3.1define*

Corollary 5.6 (rate-distortionTheorem 2′). *rate-distortionTheorem 2′*

$$\boxed{F_{1,SCX}^{\textsc{Situs}}(R_X, R_P) \leq F_{1,base} + C_F \cdot r\left(\delta(R_X) + \frac{2\varepsilon_{\mathrm{PE}}(R_P)}{C_F^2}\right)} \tag{32}$$

where$R_X R_P$StatusX encoding$\mathrm{PE}(P)$ $R_X R_P$***-Pareto***

Proof. theorem5.2define5.5 $X\mathrm{PE}(P)$Bottleneckupper boundMutual Information Mutual Information≤
□ □

[Rigorous Proof] ProofStatus:$U_X \perp\!\!\!\perp U_P \mid (X, \mathrm{PE}(P))$**assumption** otherwise$U = (U_X, U_P)$compression $R_X R_P$——Bottleneck

5.5 Theorem 3

rate-distortionTheorem 3-indistinguishabilityPerspective

Proposition 5.7 (Theorem 3rate-distortion). *Theorem 3indistinguishabilityawaiting$W_A W_B$ Rcompression$Q$sat*

$$\max_{W\in\{W_A,W_B\}} P_W(\mathit{error} \mid Q) \geq \frac{1}{2} - \sqrt{\frac{1}{2}\mathrm{D_{KL}}(P_{Q|W_A} \| P_{Q|W_B})} \tag{33}$$

$P_{Q|W_A} = P_{Q|W_B}$ *lower bound*$\geq 1/2$——

Proof. Le CamlemmacompressionKLTV ≥ 1thereforemax $\geq 1/2$ □ □

corollaryalthoughPE$R \to \infty$indistinguishabilityProposition4.1 $RW_A W_B P_{Q|W}$compression **there exists**1/2 PEtheorem4.1——

6 SCX——

6.1

CoreSCX

(1) **rate-distortion§2**SCX communicationthere existslower bound

(2) **Slepian-Wolfencoding§3encoding** communication

(3) **theorem§4**SCX optimality

(4) **Theorem 2rate-distortion§5**

Mutual Informationrate-distortion SCXtheoremTheorem 1–4 **sourceencoding**

6.2

Table 1: SCXtheoremProof

Theorem		**ProofStatus**
theorem2.1	rate-distortionlower boundasymptotic	[**Rigorous Proof**]1–3; [**Heuristic**]Compactness4
theorem2.6	asymptoticrate-distortion	[**Rigorous Proof**]
theorem3.2	SCX Slepian-Wolftheorem	[**Rigorous Proof**]
theorem3.3		[**Rigorous Proof**]
theorem4.1	SCXtheorem	[**Rigorous Proof**]
theorem4.2	theorem	[**Rigorous Proof**]; [**Heuristic**]M
theorem5.2	Theorem 2rate-distortion	[**Rigorous Proof**]
Proposition5.4		[**Heuristic**]feasibility
Proposition5.7	Theorem 3rate-distortion	[**Rigorous Proof**]

(1) **asymptotic vs. asymptotic** $M \to \infty N \to \infty n \to \infty$ SCXasymptotictheorem2.6 Scale

(2) $P(\mathbf{V})P_{Y|\mathbf{V}}$ assumptionSCX ——lower bound

(3) **Slepian-Wolf**(1)(3)§3Honest Strike

(4) **Bottleneckfeasibility**SCXMd_XIB Open Problem

6.3 SCX

This paper revealsSCX[11, 12] Open Problem

Table 2: SCX

	Status	
communication	assumption	theorem2.1lower bound
encoding	Mencoding	theorem3.2encoding
theorem	Theorem 2	theorem4.1/4.2
Theorem 2Compactness	Fano	theorem5.2rate-distortion

6.4 Open Problem

(1) **Slepian-Wolfencoding§3**SCX $n = 1$

(2) **asymptoticSCXtheorem**MtheoremVersion
Polyanskiyawaiting[8]channelencoding

(3) $\delta(R)\delta(R)$ $P_{X,Y}\delta(R)$ such that“” 95% $\delta(R) \leq \delta_0$assumption

(4) **Pareto**R_X**-**R_PStatusX encodingPE$(P)R_{\text{total}}$ $\delta(R_X)\varepsilon_{\text{PE}}(R_P)$ there existsPareto

(5) **all**Transformer
suboptimality

(6) **SCXrate-distortion-channelencoding**Theorem 4Minimax Chernoff-Steinlemmathere existsSCX—— Decays“SCXChannel Capacity”

7 Conclusion

This paper providesSCXrate-distortionSlepian-Wolfencoding ShannontheoremBottleneckCore

Core Findings:

(1) communicationSCXlower bound $P_e \geq (H(Y) - R - \log 2)/\log|\mathcal{Y}|$ Theorem 1Chernofftheorem2.1

(2) communicationSlepian-Wolfcompression reliability theorem3.3

(3) SCX“” “”assumption ——theoremtheorem4.1

(4) Theorem 2Fanorate-distortion$R = 0$ $\delta(R)$ theorem5.2

SCXtheorem Theorem 1Chernoff→ rate-distortiontheorem2.1-lower bound Theorem 2Fano→ rate-distortiontheorem5.2 Theorem 3indistinguishability→ rate-distortionProposition5.7compression Theorem 4Minimax→ Open ProblemOpen Problem6SCXChannel Capacity

asymptoticassumptionMN Slepian-Wolftheorem$M$ $\delta(R)P_{X,Y}$Fails SCX

[Honest Strike] Honest Strike The mathematical elegance of this paper SCX**Bottleneck**Shannon——communicationBottleneckSCX SCX——

Acknowledgments

—— Shannon (1948)Slepian & Wolf (1973)Wyner & Ziv (1976) Tishbyawaiting (2000)Cover & Thomas (2006) SCXCoretheoremTheorem 1–4This paper provides `theory/information_theory/`DirectorySCX Project, 2026theorem

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